

SCOPE DOCUMENT

Supplier & Purchasing Management Integration

Connecting Product, Purchasing, Inventory, and Sales into one flow — current-state assessment, gap analysis, and build approach.

Prepared for: Client — KS Shop ERP

Prepared by: Muhammad Widy Ramadhani

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System reviewed: KS Shop ERP (Go / PostgreSQL / React)

1. Executive Summary

Most of what you're asking for is **not a new build** — Supplier Management and Purchasing Management already exist in the system today as working modules. What you're asking for is an **upgrade** of both, plus connecting them properly to Product, Inventory, and Sales so the full flow (Product → Purchasing → Incoming Stock → Receiving → Physical Stock → Sales → Supplier Performance) works automatically instead of as separate, disconnected pieces.

Three parts of your request are genuinely new and don't exist in any form today: **automatic product code generation** (category prefix + sequence), the **product↔supplier relationship view** (which suppliers sold which products), and **supplier performance metrics** tied to sales.

Section 2 lays out, module by module, what exists today vs. what you're asking for. Section 3 explains how we'll close each gap. Section 4 lists decisions we need from you before work starts on certain parts — these aren't formalities, they change what gets built. Section 5 is a suggested build order. Section 6 covers scope boundaries and risk.

2. Current State vs. Requested State

2.1 Product Code

Today	You're Asking For	Gap
Product code (SKU) is a free-text field. Staff type it manually when creating a product. The actual sequence is tracked outside the system, in a separate Google Sheet.	Code is auto-generated as Category Prefix + Sequential Number (e.g. next Pants after P087 is P088), with its own counter per category.	No auto-generation logic exists. Categories have no "prefix" concept yet. The Sheet's current numbering isn't in the system, so the counter needs to be seeded from it.

2.2 New Product Flow

Today	You're Asking For	Gap
A product is created in Product Management with a manually typed code. Purchasing lets staff select any existing product/variant — there's no rule preventing a brand-new SKU from being typed directly into a purchase.	New products can <i>only</i> be created in Product Management (with the code auto-generated there). Purchasing only ever selects from existing products — never creates one.	Purchasing's product picker needs to become select-only, with no "create new" path, and needs a proper search/picker instead of the current flat dropdown of every variant.

2.3 Supplier Management

Today	You're Asking For	Gap
Fields: Name, Contact Person, Phone, Address, Active/Inactive. Basic list + CRUD, no filtering beyond active/inactive.	Supplier ID, Name, Contact Person, WeChat/Contact, Category, Style, Price Range, Notes, Attachments, Status — with Category/Style/Price Range as filterable tags, built to comfortably handle ~300 suppliers.	Need a Supplier ID field, WeChat field, file attachments, and a tagging system for Category/Style/Price Range with filter UI. The list view needs to support tag-based filtering at scale.

2.4 New Supplier During Purchasing

Today	You're Asking For	Gap
Doesn't exist. A purchase can only select a supplier that's already fully created in Supplier Management, from a separate page.	A "+ Add New Supplier" option inline in the purchase form — Name + WeChat is enough to proceed. The purchase isn't blocked. The new supplier shows up in Supplier Management marked "Incomplete," with a reminder to finish the profile later.	Fully new: inline quick-create modal, an "Incomplete" status state, and a way to surface/remind on incomplete suppliers.

2.5 Purchasing Management

Today	You're Asking For	Gap
Fields: Supplier, Order Date, Status (Waiting/Received only), Notes, line items (variant, qty, unit cost). No ETA, no Purchase ID, no attachments, no partial receiving — a purchase is received all-or-nothing.	Purchase ID, Supplier, Order Date, ETA, Status, Notes, Attachments (WeChat/payment/product screenshots), and per-line Product/SKU, Color, Size, Quantity, Unit Cost, Total Cost — with partial receiving supported.	New fields (ETA, human-readable Purchase ID, Total Cost), attachment support (current upload endpoint only accepts a single image, not arbitrary screenshots/files), a proper multi-status flow (Waiting → Partially Received → Received), and partial-quantity receiving logic.

2.6 Product ↔ Supplier Relationship

Today	You're Asking For	Gap
No relationship tracked at all between products and suppliers.	Automatic, derived from actual purchase history — no manual linking. Product page shows which suppliers have supplied it; Supplier page shows which products have been purchased from them.	No new tables needed — this is a reporting view built by joining existing purchase records. Needs a new tab/section on both the Product and Supplier detail pages.

2.7 Purchasing ↔ Inventory

Today	You're Asking For	Gap
"Incoming Stock" exists as a number on each product variant, but it's manually typed by staff — it has no connection to actual purchase orders. Receiving a purchase today adds the full quantity straight to Physical (Available) Stock in one shot; there's no partial receiving.	Creating a purchase automatically adds the quantity to Incoming Stock. Confirming receipt moves received quantity from Incoming Stock into Physical Stock — partial receiving supported (e.g. ordered 200, received 120 → Physical +120, Incoming stays at 80).	This is the most structurally significant change. Incoming Stock needs to stop being a manually-edited field and become purchase-driven. See the risk callout in Section 6 — Incoming Stock currently also counts toward what customers can already buy/pick, before goods arrive.

2.8 Supplier Performance

Today	You're Asking For	Gap
Doesn't exist. Sales reporting (GMV, quantity sold) exists at the product/category level but has no supplier dimension.	Per supplier: Total Purchase Amount, Total Purchase Quantity, Number of Purchases, Incoming Quantity, Quantity Sold, GMV, NGR — all calculated automatically from purchasing + sales data.	Purchase-side metrics are straightforward once purchasing is properly tracked. Sales-side metrics require a rule for attributing a sale to a supplier when a product has multiple suppliers (see Decision #1). "NGR" also isn't a metric defined anywhere in the current system (see Decision #2).

3. How We'll Build It

Product Code Generator **NEW**

- Add a code prefix to each category (e.g. Clothes → C, Pants → P) and a running counter per category.
- On product creation, the system generates the next code server-side and shows it read-only — no manual typing, no collisions when two staff create products at the same time.
- Existing products keep their current codes untouched; the counters are seeded from the highest number already in use per category (ideally imported from your current Google Sheet — see Decision #6).

Purchasing Product Picker **EXTEND**

- Replace the current flat "every variant in one dropdown" selector with a proper searchable product/variant picker (an existing picker component elsewhere in the system can be reused).
- Remove any path to typing a brand-new SKU from inside Purchasing — selection-only, by design.

Supplier Management: Fields, Tags, Attachments **EXTEND**

- Add Supplier ID (auto-numbered display code), WeChat/Contact, Notes, Attachments, and a Status field (Complete / Incomplete).
- Add Category, Style, and Price Range as taggable, filterable attributes — built so filtering stays fast at ~300 suppliers.
- Extend the file-upload capability so a supplier record can hold multiple attachments (current upload handler is limited to one image at a time — needs broadening to general files/screenshots).

Inline "+ Add New Supplier" **NEW**

- A lightweight create-supplier modal inside the purchase form, capturing just Name and WeChat/Contact.
- New supplier is saved immediately with Status = Incomplete and appears in Supplier Management right away; the purchase itself is never blocked by an incomplete profile.
- A filter/reminder surface in Supplier Management for "Incomplete" suppliers still needing full details.

Purchasing: ETA, Purchase ID, Attachments, Partial Receiving **EXTEND**

- Add ETA and a human-readable Purchase ID to each purchase record; compute and display Total Cost per line and per purchase.
- Attachments support for order/payment/product screenshots (same upload broadening as Supplier attachments).

- Add a "received quantity" per line item so a purchase can move through Waiting → Partially Received → Received, instead of all-or-nothing.

Purchasing ↔ Inventory Wiring EXTEND

- Creating a purchase adds its quantities to Incoming Stock automatically; Incoming Stock stops being a manually-typed field once this is live.
- Confirming receipt (full or partial) moves the received amount from Incoming Stock to Physical/Available Stock, leaving any un-received remainder in Incoming Stock.
- Depends on Decision #3 (whether incoming/unreceived stock should still count as sellable, as it does today) and Decision #7 (how cost price is set when a product has multiple suppliers or partial receipts).

Product ↔ Supplier View NEW

- No new data storage — a read-only view built by joining purchase history back to products and suppliers.
- New "Suppliers" tab on the Product detail page; new "Products Purchased" tab on the Supplier detail page.

Supplier Performance Dashboard NEW

- Purchase-side metrics (Total Purchase Amount/Quantity, Number of Purchases, Incoming Quantity) — computed directly from purchase records, straightforward once the above is in place.
- Sales-side metrics (Quantity Sold, GMV, NGR) — require Decision #1 (attribution rule for multi-supplier products) and Decision #2 (NGR formula) before this can be built.

4. Decisions Needed From You

These aren't just FYI — each one changes what actually gets built. We'd like answers before starting the phases that depend on them (flagged in Section 5).

1. When a product has multiple suppliers, how should a sale be attributed to a supplier for performance reporting?

Why it matters: crediting every supplier who ever supplied the product double-counts sales; splitting by purchase-quantity share is simpler but approximate; tracking by lot/batch (FIFO) is the most accurate but the most work to build.

2. What's the exact formula for NGR (Net Gross Revenue)?

Why it matters: this term doesn't exist anywhere in the current system. We need to know what it's net of — returns, discounts, shipping cost, platform fees — before it can be calculated.

3. Should stock that's still "incoming" (ordered but not yet received) remain sellable/pickable, as it does today?

Why it matters: right now, Incoming Stock already counts toward what staff can sell and pick. Wiring Purchasing into Incoming Stock without addressing this would let customers buy stock that's still on a boat.

4. Is the new Product Code a product-level code only, separate from each variant's own SKU (e.g. P088 the product, P088-BLACK-M the variant)?

Why it matters: confirms we're not changing how variant SKUs already work in orders/picking — only adding the product-level code layer on top.

5. Code format details: zero-padding (P088 vs P88), what happens if two categories would share a prefix, and what happens past 999 (P999 → P1000)?

6. Can you share the current Google Sheet so we can seed each category's counter from the real current numbers?

Why it matters: without this, counters start from zero and will collide with codes already in use.

7. When a product is purchased from multiple suppliers, or a purchase is only partially received, how should the product's cost price be updated?

Why it matters: today, receiving a purchase simply overwrites the cost price with the latest unit cost. That breaks down once partial receiving and multiple suppliers are both live — we need a rule (last cost, weighted average, etc.) since cost price feeds your existing profit reporting.

8. Should over-receiving (receiving more than was ordered) be allowed, and can a purchase be edited or cancelled after it's already been (partially) received?

5. Recommended Build Order

Phase	Covers	Depends On
Phase 1	Product code auto-generation (category prefixes, sequence counters, seeding from existing codes)	Decisions 4, 5, 6
Phase 2	Supplier Management upgrade — new fields, tags/filters, attachments, inline "+ Add New Supplier" during purchasing	None — can start independently
Phase 3	Purchasing upgrade — ETA, Purchase ID, attachments, proper product picker, partial receiving, incoming-stock wiring	Decisions 3, 7, 8
Phase 4	Product ↔ Supplier relationship views (derived from purchase history)	Phase 3 complete (needs real purchase data flowing)
Phase 5	Supplier Performance dashboard (purchase + sales metrics, GMV, NGR)	Decisions 1, 2; Phases 3 & 4 complete

6. Scope Boundaries & Risk

Timeline and cost are not in this document

Effort, timeline, and cost estimates are intentionally left out of this scope document. Producing a reliable estimate requires time to assess how each change interacts with the live system (especially the Purchasing ↔ Inventory wiring below) — that assessment is its own step, done once the decisions in Section 4 are answered, and will follow as a separate estimate.

Changing Incoming Stock behavior touches live order flow

Incoming Stock isn't just a display number today — it's already part of the calculation that decides what staff can sell and pick right now, before goods physically arrive. Wiring Purchasing into it (Section 2.7) is the highest-impact change in this scope and needs Decision #3 answered before it's built, to avoid unintentionally changing what's currently sellable.

Product code sequence depends on the Google Sheet handoff

If the current Sheet isn't provided (Decision #6), auto-generated codes will start from scratch per category and are very likely to collide with codes already assigned manually. This needs to happen before Phase 1 ships.